Quick Syntax Cheat Sheet

Converted from the supplied TXT source; content preserved.

CSE 4165 WEB PROGRAMMING - ONE-PAGE-STYLE CHEAT SHEET

JAVASCRIPT

```
let total = 0, count = 0;  // outside function for state
function run() {
    let x = Number(document.getElementById("input").value);
```

total += x;

count++;

```
    if (total <= 5) { message = "A"; }
```

else if (total <= 12) { message = "B"; }

else { message = "C"; }

document.getElementById("out").innerText = message;

}

PHP FORM

<form method="post">

Number: <input type="number" name="x">

<button name="btn">Calculate</button>

</form>

```
<?php
if (isset($_POST["btn"])) {
```

$x = (float)$_POST["x"];

echo $x;

}

?>

PHP FUNCTION

```
function calculate($people, $capacity) {
```

$units = ceil($people / $capacity);

$left = $units * $capacity - $people;

return $left;

}

MYSQLI LOOP

$conn = new mysqli("localhost", "root", "", "dbname");

```
if ($conn->connect_error) { die("Connection failed: " . $conn->connect_error); }
```

$res = $conn->query("SELECT name AS n FROM table_name");

```
while ($row = $res->fetch_assoc()) { echo $row["n"] . "<br>"; }
```

$conn->close();

SQL CLAUSE ORDER

```
SELECT columns, COUNT(*) AS total
```

FROM table_name

WHERE row_condition

GROUP BY group_column

HAVING COUNT(*) > 1

ORDER BY total DESC

LIMIT 2;

CAP UPDATE

```
UPDATE table_name
```

SET score = score + 5

WHERE score > 80 AND score + 5 <= 90;

FORMULAS

Needed = number of people * units per person

Whole units = ceil(needed / capacity)

Leftover = whole units * capacity - needed

Waste = leftover * price per unit

Best two CT average = (ct1 + ct2 + ct3 - min(ct1,ct2,ct3)) / 2

Average speed for same-distance up/down = total distance / total time

COMMON TRAPS

- ID in HTML/JS; name in PHP/POST.

- Cast input before numeric calculations.

- Put JS totals/counters outside functions.

- Apply special warning after ordinary feedback.

- ceil for whole venues/screens/pizzas.

- WHERE filters rows; HAVING filters groups.

- Test the updated value when a rule says “only if the result <= cap”.

- Alias SQL aggregates with AS.

- Avoid duplicate HTML IDs.

- Re-import setup.sql after UPDATE tests.

- Mark contradictions: Spring 2024 digit product vs sample sum; Fall 2025 Excellent threshold prose vs sample; Spring 2025/Summer 2025 fixed cost vs waste sample.